

Quiz 6 - Decidability & Reducibility

Date: December 14, 2024

Name: _____

ERP ID: _____

Total Marks: 10 + 5 bonus points

Question 1

[5 Points]

Let $A = \{ \langle M \rangle \mid M \text{ is a DFA that doesn't accept any string containing an odd number of 1s} \}$. Show that A is decidable.

Question 2**[5 Points]**

Show that EQ_{TM} is not Turing-recognizable.

Bonus Question**[5 Points]**

Let $T = \{(i, j, k) \mid i, j, k \in \mathbb{N}\}$. Show that T is countable.